

Autodesk® Scaleform®

Scaleform 3D

Scaleform 3.2

3D

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2.03
2012 5 9

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Autodesk® Scaleform® 4.3

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Autodesk Scaleform

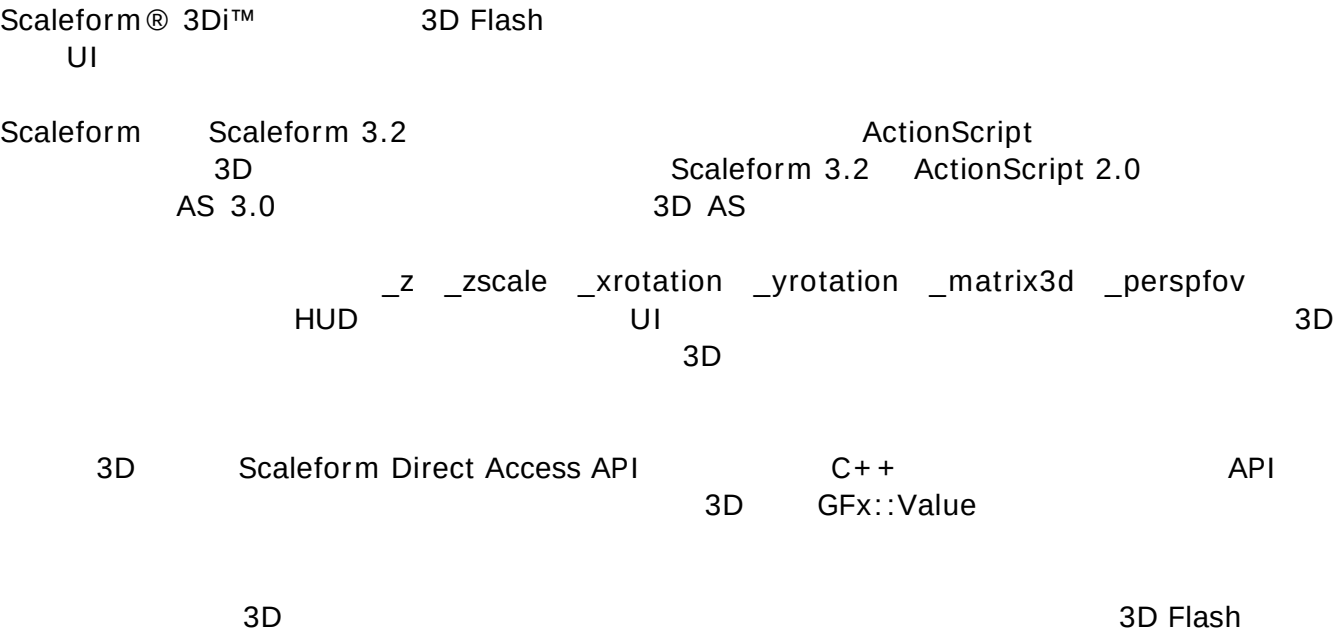
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	Greenbelt, MD 20770, USA
	www.scaleform.com
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	+1 (301) 446-3200
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1 概要



2 Flash 10 との類似性

Scaleform m

3Di 3Di Scaleform

Flash 10 3D

AS2

2.1 制約事項

Flash 10 3D

3Di

- 1.
- 2.

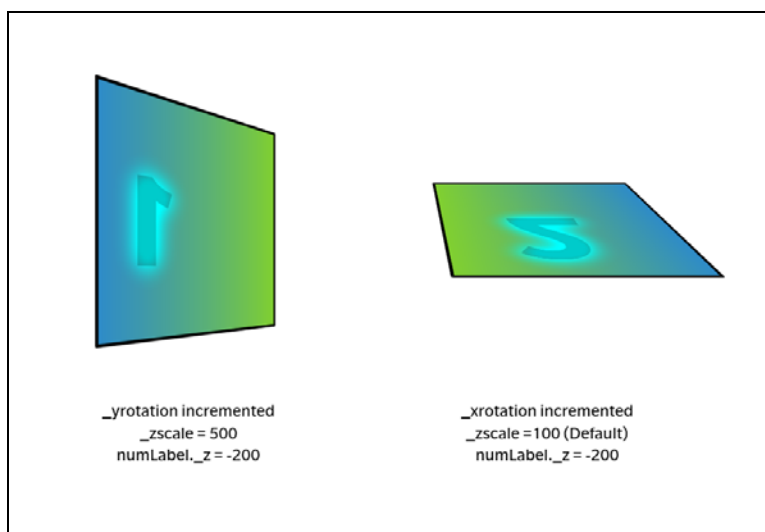


図 1: カリングされずにオブジェクトの後ろ側で描画される裏向き数字の例

2.2 実装

Scaleform 3Di

Flash 10 3D

2D 3D

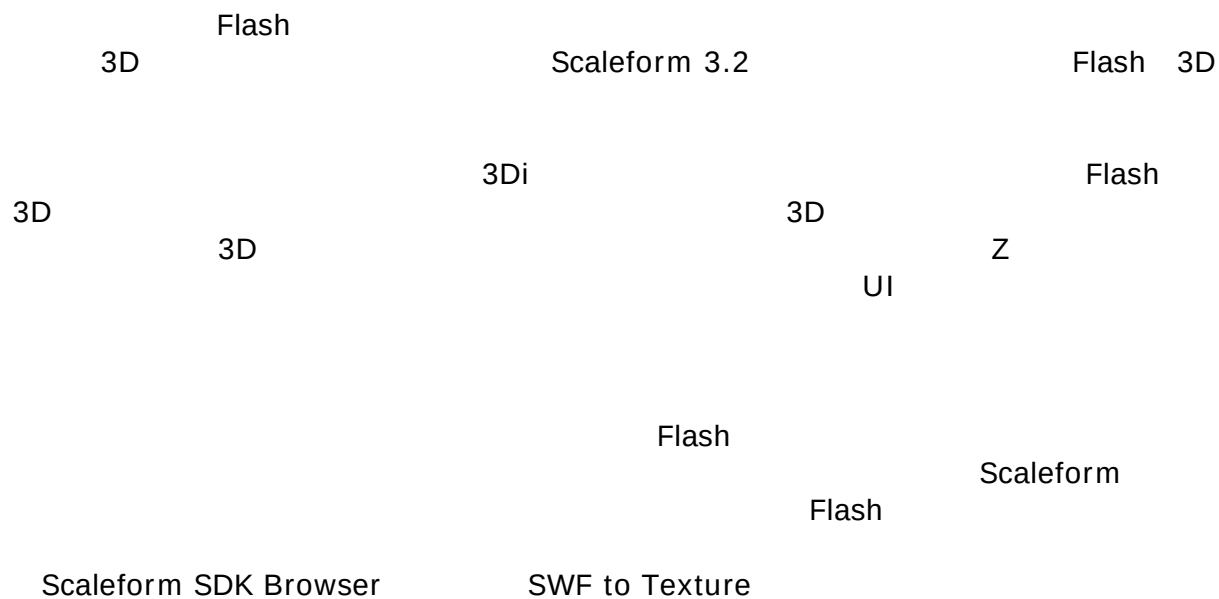
2D
3D
Flash 10

3D

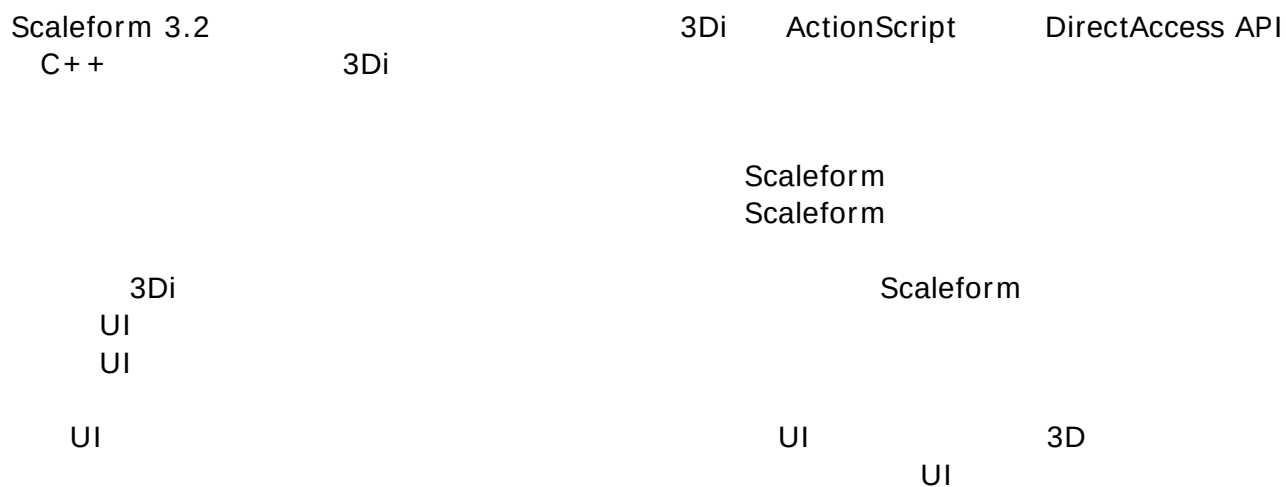
3 3DでのFlashの表示

Scaleform 3.2 Flash 3D 2

3.1 レンダーからテクスチャへの変換



3.2 3Di



4 3Di API

4.1 ActionScript 2 API

3DiFlash3

23Dx3y3z3Dnull3D

ActionScriptzfoo._matrix3d = null03D2D

2Dz33D_xscaleActionScriptx y

3DScaleformFlash 10FOV

+X55°+Y+Z

3DFlash

- Movie clips
- Text fields
- Buttons

3DiActionScript

- **_z** - Z0
- **_zscale** - Z100
- **_xrotation** - X0
- **_yrotation** - Y0
- **_matrix3d** - 164×43DNULL3D
- **_perspfov** - 2D0180Field Of View-155°FOV

```
gfxExtensions    true
```

3Di

Y 45

$$X \qquad \qquad \qquad Z$$

3D

```
function PixelsToTwips(iPixels):Number
{
    return iPixels*20;
}
function TranslationMatrix(tX, tY, tZ):Array
{
    var matrixC:Array = [    1, 0, 0, 0,
                             0, 1, 0, 0,
                             0, 0, 1, 0,
                             tX,tY,tZ,1];
}
```

```

        return matrixC;
    }

    this._matrix3d = TranslationMatrix(PixelsToTwips(100),PixelsToTwips(100),0);

```

4.2 C++から 3Di を利用する

4.2.1 Matrix4x4

4×4 row major Matrix4x4

3D
Render_Matrix4x4.h

4.2.2 Direct Access

3Di API Gfx::Value 3D Direct Access 3Di

ActionScript

Gfx_Player.h [Gfx::Value::DisplayInfo](#)

```

void    SetZ(Double z)
void    SetXRotation(Double degrees)
void    SetYRotation(Double degrees)
void    SetZScale(Double zscale)
void    SetFOV(Double fov)
void    SetProjectionMatrix3D(const Matrix4F *pmat)
void    SetViewMatrix3D(const Matrix3F *pmat)
bool    SetMatrix3D(const Render::Matrix3F& mat)

```

```

Double  GetZ() const
Double  GetXRotation() const
Double  GetYRotation() const
Double  GetZScale() const
Double  GetFOV() const
const Matrix4F* GetProjectionMatrix3D() const
const Matrix3F* GetViewMatrix3D() const
bool    GetMatrix3D(Render::Matrix3F* pmat) const

```

4.2.3 Render::TreeNode

3D

Render::TreeNode

Gfx_Player.h

```
void    SetProjectionMatrix3D(const Matrix4F& m)
void    SetViewMatrix3D(const Matrix3F& m);
```

4.2.4例 : Direct Access API を使ったパースペクティブ FOV の変更

```
Ptr<Gfx::Movie> pMovie = ...;
Gfx::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "_root.Window");
if (bOK)
{
    Gfx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        // set perspectiveFOV to 150 degrees
        Double oldPersp = dinfo.GetFOV();
        dinfo.SetFOV(150);
        tmpVal.SetDisplayInfo(dinfo);
    }
}
```

5 パースペクティブ

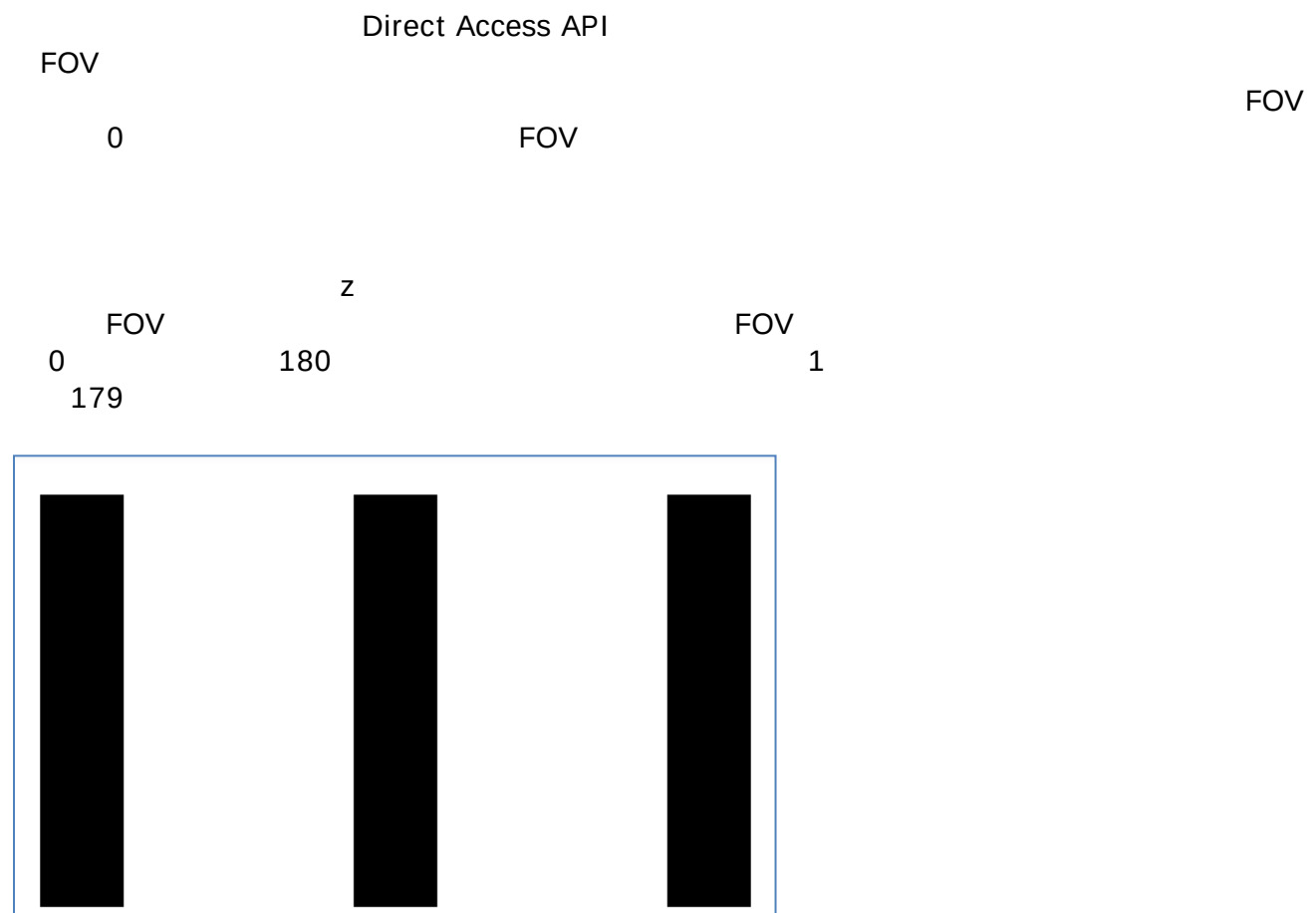


図 2: オリジナルの（回転していない）Flash イメージ

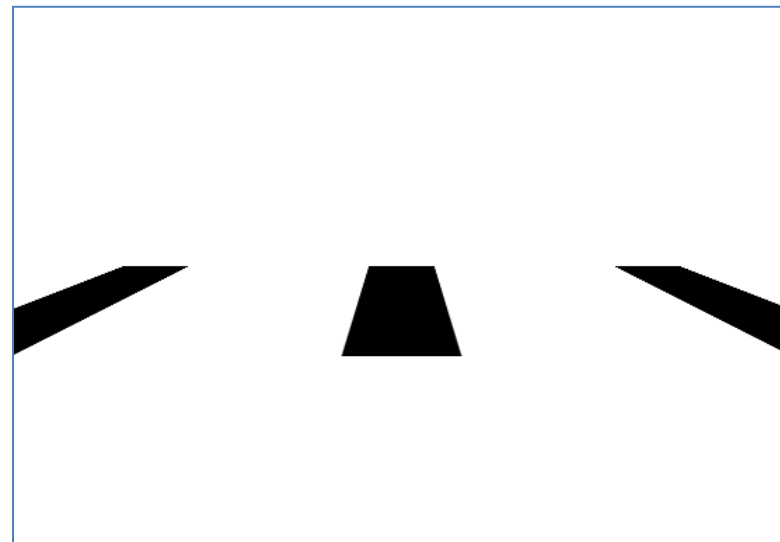


図 3: X 方向に-80 の回転を与え、パースペクティブはデフォルト（55°）

orthographic projection
 ActionScript property _perspfov -1

C++ Direct Access API

// 1. Perspective FOV -1

```
GfX::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root"); //"_root" for AS2
if (bOK)
{
  GfX::Value::DisplayInfo dinfo;
  bOK = tmpVal.GetDisplayInfo(&dinfo);
  if (bOK)
  {
    dinfo.SetPerspFOV(-1);
    tmpVal.SetDisplayInfo(dinfo);
  }
}
```

// 2.
 void MakeOrthogProj(const RectF &visFrameRectInTwips, Matrix4F *matPersp)

```
{
  const float nearZ = 1;
  const float farZ = 100000;
  float DisplayHeight = fabsf(visFrameRectInTwips.Height());
  float DisplayWidth = fabsf(visFrameRectInTwips.Width());
  matPersp->OrthoRH(DisplayWidth, DisplayHeight, nearZ, farZ);
}
GfX::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "root.mc");
if (bOK)
{
  GfX::Value::DisplayInfo dinfo;
  bOK = tmpVal.GetDisplayInfo(&dinfo);
  if (bOK)
  {
    Matrix4F perspMat;
    MakeOrthogProj(PixelsToTwips (pMovie->GetVisibleFrameRect()), &perspMat);
    dinfo.SetProjectionMatrix3D(&perspMat);
    tmpVal.SetDisplayInfo(dinfo);
  }
}
```

Projection

Ptr<Movie> pMovie = ...

```
// compute ortho matrix
Render::Matrix4F ortho;
const RectF &visFrameRectInTwips = PixelsToTwips(pMovie->GetVisibleFrameRect());
const float nearZ = 1;
const float farZ = 100000;
ortho.OrthoRH(fabs(visFrameRectInTwips.Width()), fabs(visFrameRectInTwips.Height()),
nearZ, farZ);
pMovie->SetProjectionMatrix3D(ortho);
```

5.1 AS3 でのパースペクティブの設定

ActionScript 3 PerspectiveProjection

Example:

```
import flash.display.Sprite;

var par:Sprite = new Sprite();
par.graphics.beginFill(0xFFCC00);
par.graphics.drawRect(0, 0, 100, 100);
par.x = 100;
par.y = 100;
par.rotationX = 20;

addChild(par);

var chRed:Sprite = new Sprite();
chRed.graphics.beginFill(0xFF0000);
chRed.graphics.drawRect(0, 0, 100, 100);
chRed.x = 50;
chRed.y = 50;
chRed.z = 0;

par.addChild(chRed);

var pp3:PerspectiveProjection=new PerspectiveProjection();
pp3.fieldOfView=120;
par.transform.perspectiveProjection = pp3;
```

6 3D での立体画像

Scaleform 3Di 3D UI 2

Scaleform 3Di NVIDIA 3D Vision Kit 3D
NVIDIA

Scaleform Display 1 2

P(c)-0 -13(V)-3(ID)-2(IA)TJC2_1 1 3Tf0 Tc 0 Tw 70929090E11096B097 -1.272 E1109690640061A116

3D Vision

3D Vision
CTRL F4

IR

CTRL F3

Scaleform Player

CTRL F5 CTRL F6

*Set up stereoscopic 3D
advanced in-game settings*

NVIDIA
Set Keyboard Shortcuts

Stereoscopic 3D
Enable the
CTRL F5 CTRL F6
CTRL F7

CTRL F5

10

15

CTRL F5 CTRL F6

CTRL F7

NVIDIA

3D
3D

API

API

<http://developer.nvidia.com/object/nvapi.html>.

http://developer.download.nvidia.com/presentations/2009/SIGGRAPH/3DVision_Develop_Design_Play_in_3D_Stereo.pdf.

6.2 Scaleform ステレオ API



6.2.1

Scaleform m

Scaleform 4

Render::Viewport::SetStereoViewport

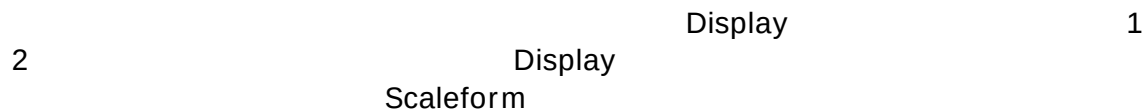
SF4

```
//
View_Stereo_SplitV      = 0x40,
View_Stereo_SplitH      = 0x80,
View_Stereo_AnySplit    = 0xc0,

void SetStereoViewport(unsigned display);
```

6.2.1 Scaleform Stereo 初期化

```
Render::StereoParams s3DInfo;
s3DInfo.DisplayDiagInches = 46;           // 46 inch TV
s3DInfo.DisplayAspectRatio = 16.f / 9.f;
s3DInfo.Distortion = .75;                 // 0 to 1
s3DInfo.EyeSeparationCm = 6.4;           // this will default 6.4cm
pRenderHAL->SetStereoParams(s3DInfo);
```



```
pMovie->Advance(delta);

pRenderer->GetHAL()->SetStereoDisplay(Render::StereoLeft);
pMovie->Display();

pRenderer->GetHAL()->SetStereoDisplay(Render::StereoRight);
pMovie->Display();
```

7 サンプルファイル

3Di
FLA

Scaleform SDK

3D

Flash

SWF

C:\Program Files (x86)\Scaleform\GFx SDK 4.2\Bin\Data\AS2or AS3\Samples



図 4: 3DGenerator

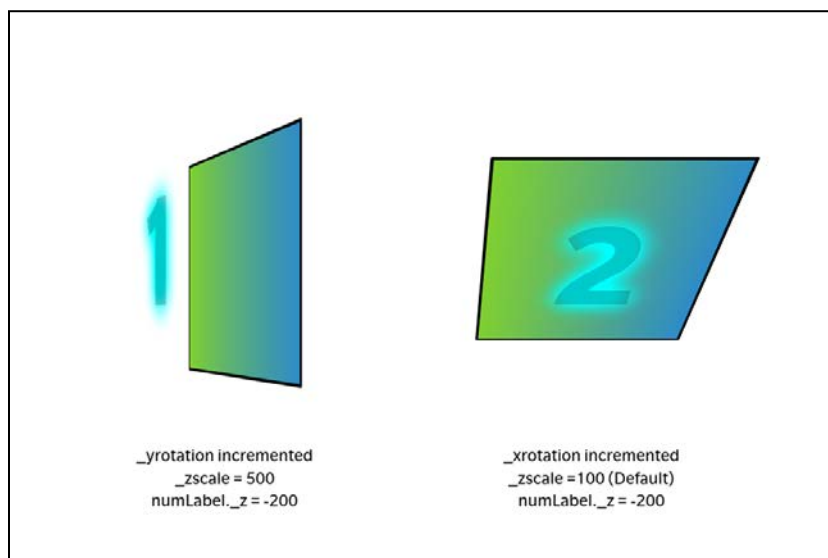


図 5: 3D スクエア



図 6: 3D ウィンドウ